

UNDER A SIMPLICITY HYPOTHESIS, THE ORDER OF A [3, 1; 4]-MIXED CAGE IS TWELVE

ABSTRACT. Araujo-Pardo, Hernández-Cruz and Montellano-Ballesteros proved $n[z, 1; 4] \leq 3(z + 1)$ for odd z and $n[z, 1; 4] \leq 3z + 2$ for even z , with equality for $z \in \{1, 2\}$, and proposed as a first problem the search for a matching lower bound. Under the standing hypothesis that mixed graphs are *simple*, we compute the single case $z = 3$: $n[3, 1; 4] = 12$. The upper half is their own circulant on $\mathbb{Z}_{12}$; the lower half is two lines from an elementary digraph lemma — a loopless, digon-free, directed-triangle-free digraph on m vertices with minimum out-degree $\delta \geq 1$ has $m \geq 2\delta + 2$ — applied to the digraph induced on the out-neighbourhood of one vertex. The same argument gives $n[z, 1; 4] \geq \lceil (5z + 6)/2 \rceil$ for all $z \geq 2$. The simplicity reading is a hypothesis, not a theorem about the source, whose text contains both a sentence allowing multiple edges and arcs and a definition requiring a simple graph; we exhibit an order-6 multigraph object for which the value 12 fails, so the multigraph variant is not decided here.

1. THE PROBLEM

A *mixed graph* has both edges and arcs. Following Araujo-Pardo, Hernández-Cruz and Montellano-Ballesteros [1], a mixed graph is $[z, r]$ -*regular* if every vertex is the tail of exactly z arcs, the head of exactly z arcs, and incident with exactly r edges; a *walk* traverses an edge in either direction and an arc only forwards; the *girth* is the length of a shortest cycle in that sense; and $n[z, r; g]$ denotes the least order of a $[z, r]$ -regular mixed graph of girth g , a $[z, r; g]$ -*mixed cage*. Throughout, mixed graphs are *simple*: at most one arc joins a given ordered pair of vertices, and no arc is parallel to an edge. This convention is a hypothesis, and it is load-bearing; we return to it below. Since $r = 1$ means the edges form a perfect matching, the order of a $[z, 1; g]$ -mixed graph is even.

The target statement is Theorem 3.3 of [1], verbatim:

Theorem. Let $z \geq 1$ be an integer, then we have that $n[z, 1; 4] \leq 3(z + 1)$ if z is odd, $3z + 2$ otherwise. Moreover, the equality holds for $z \in \{1, 2\}$.

The equality clause covers $z \in \{1, 2\}$; the case treated here is $z = 3$, where $3(z + 1) = 12$. A matching lower bound, in general and not specifically at $z = 3$, is asked for in the closing paragraph of Section 4 of [1]:

2020 *Mathematics Subject Classification.* 05C35, 05C20, 05C38.

Key words and phrases. mixed graph, mixed cage, girth, directed cage, circulant.

As a first problem, we propose to find a lower bound for the order of a $[z, 1; 4]$ -mixed cage. This, together with the upper bound given in Theorem 3.3, could result in an exact value for the order of this family of mixed cages.

The simplicity convention. Two clauses of [1] conflict: its opening says “We allow multiple edges and arcs”, while its definition of a mixed regular graph asks for “a simple and finite graph G ”. We do not resolve which reading the problem quoted above intends, and we make no claim to settle or extend that problem. Instead we prove a simple-graph statement, with simplicity as an explicit hypothesis; the remark in Section 3 shows that the value below is false under the multigraph reading.

Theorem 1. *For simple mixed graphs, $n[3, 1; 4] = 12$.*

Consequently, for simple mixed graphs, the odd- z bound $3(z + 1)$ of Theorem 3.3 of [1] is tight at $z = 3$. Whether this is the same assertion as an extension of that theorem’s equality clause depends on the unresolved convention discussed above.

2. THE WITNESS

The inequality $n[3, 1; 4] \leq 12$ is Lemma 3.1(1) of [1] at $n = 3$: the mixed circulant C on $\mathbb{Z}_{12}$ with

arcs $i \rightarrow i + 1, i \rightarrow i + 2, i \rightarrow i + 7$ and edges $\{i, i + 6\}$,

all indices modulo 12. In full:

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a 0 1  a 0 2  a 0 7  a 1 2  a 1 3  a 1 8  a 2 3  a 2 4  a 2 9
a 3 4  a 3 5  a 3 10 a 4 5  a 4 6  a 4 11 a 5 0  a 5 6  a 5 7
a 6 1  a 6 7  a 6 8  a 7 2  a 7 8  a 7 9  a 8 3  a 8 9  a 8 10
a 9 4  a 9 10 a 9 11 a 10 0 a 10 5 a 10 11 a 11 0 a 11 1 a 11 6
e 0 6  e 1 7  e 2 8  e 3 9  e 4 10 e 5 11

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(a $u\ v$ is the arc $u \rightarrow v$; e $u\ v$ is the edge $\{u, v\}$; 36 arcs and 6 edges on 12 vertices.)

Out-degree and in-degree are 3, the in-differences being $12 - 1 = 11$, $12 - 2 = 10$, $12 - 7 = 5$. The edges form a perfect matching, so $r = 1$. There is no digon, since none of 11, 10, 5 lies in $\{1, 2, 7\}$; no arc is parallel to an edge, since $6 \notin \{1, 2, 7\}$. There is no directed triangle, because the ten three-term sums from $\{1, 2, 7\}$ with repetition are

$$3, 4, 9, 5, 10, 3, 6, 11, 4, 9 \pmod{12},$$

none of them 0; and no mixed triangle (an edge $\{x, y\}$ together with arcs $x \rightarrow w \rightarrow y$), because the six two-term sums

$$2, 3, 8, 4, 9, 2 \pmod{12}$$

avoid the edge difference 6. Finally $1 + 2 + 2 + 7 = 12$, so $0 \rightarrow 1 \rightarrow 3 \rightarrow 5 \rightarrow 0$ is a directed 4-cycle. Hence the girth is exactly 4 and C is a $[3, 1; 4]$ -mixed graph of order 12.

3. THE LOWER BOUND

Everything rests on one elementary lemma about digraphs.

Lemma 2. *Let H be a digraph on m vertices with no loop, no digon, no directed triangle, and minimum out-degree $\delta \geq 1$. Then $m \geq 2\delta + 2$.*

Proof. The number of arcs is at least $m\delta$ and equals the sum of the in-degrees, so some vertex a has in-degree at least δ . Put $Q = N^-(a)$ and $P = N^+(a)$; then $|Q|, |P| \geq \delta \geq 1$, and $P \cap Q = \emptyset$ because a vertex in both would give a digon at a , and $a \notin P \cup Q$ because there is no loop. Pick $p \in P$. Then $N^+(p)$ omits p (no loop), omits a (else $a \rightarrow p \rightarrow a$ is a digon), and omits every $q \in Q$ (else $a \rightarrow p \rightarrow q \rightarrow a$ is a directed triangle on three distinct vertices). Hence

$$\delta \leq |N^+(p)| \leq m - 2 - |Q| \leq m - 2 - \delta. \quad \square$$

Lemma 2 is tight at $\delta = 1$: the directed 4-cycle has $m = 4 = 2 \cdot 1 + 2$. That is the case used below. At $m = 3$ the lemma says there is no such digraph with $\delta \geq 1$, and one sees it directly: follow $a \rightarrow b \rightarrow c$; then $c \rightarrow a$ is a directed triangle and $c \rightarrow b$ a digon, so c has nowhere to point.

Theorem 3. *For every $z \geq 2$, $n[z, 1; 4] \geq \lceil (5z+6)/2 \rceil$, and hence $n[z, 1; 4] \geq 2\lceil (5z+6)/4 \rceil$ because $r = 1$ forces the order to be even.*

Proof. Let (D, M) be a $[z, 1; 4]$ -mixed graph of order n , with M the perfect matching of edges. Fix a vertex u , let $v = M(u)$, $A = N^+(u)$, $B = N^-(u)$ and $S = V \setminus (\{u, v\} \cup A \cup B)$. These four sets are pairwise disjoint: $A \cap B = \emptyset$ since a common vertex would give a digon, $u \notin A \cup B$ since there is no loop, and $v \notin A \cup B$ since an arc parallel to the edge $\{u, v\}$ is a cycle of length 2. As $|A| = |B| = z$, we get $|S| = n - 2z - 2$.

Let $a \in A$. Then $N^+(a)$ omits u (a digon with $u \rightarrow a$), omits v (otherwise $u \rightarrow a \rightarrow v$ together with the edge $\{v, u\}$ is a cycle of length 3), and omits every $b \in B$ (otherwise $u \rightarrow a \rightarrow b \rightarrow u$ is a directed triangle). So $N^+(a) \subseteq (A \setminus \{a\}) \cup S$, whence $|N^+(a) \cap A| \geq z - |S|$.

Let H be the digraph induced on A . It has $m = |A| = z$ vertices, and it inherits from D the absence of loops, digons and directed triangles. If $z - |S| \leq 0$ then $|S| \geq z$ and $n = 2z + 2 + |S| \geq 3z + 2$. Otherwise H has minimum out-degree at least $z - |S| \geq 1$, so Lemma 2 gives $z \geq 2(z - |S|) + 2$, i.e. $|S| \geq \lceil (z + 2)/2 \rceil$ and therefore

$$n = 2z + 2 + |S| \geq 2z + 2 + \left\lceil \frac{z+2}{2} \right\rceil = \left\lceil \frac{5z+6}{2} \right\rceil.$$

For $z \geq 2$ the second alternative is the smaller of the two, so it is the bound. The parity refinement is immediate. $\square$

Proof of Theorem 1. Suppose a $[3, 1; 4]$ -mixed graph has order 10. Then, in the notation of the previous proof, $|S| = 10 - 2 \cdot 3 - 2 = 2$, so the digraph induced on $A = N^+(u)$ has 3 vertices, no loop, no digon, no directed triangle and minimum out-degree at least $3 - 2 = 1$. Lemma 2 demands $3 \geq 4$, a

contradiction. Orders below 10 are excluded by the same one-vertex count: $n \leq 6$ is impossible because $\{u\} \cup \{v\} \cup A \cup B$ already has $2+3+3 = 8$ elements, and $n = 8$ forces $|S| = 0$, so H would have minimum out-degree 3 on 3 vertices and Lemma 2 demands $3 \geq 8$. As the order is even, $n[3, 1; 4] \geq 12$; equivalently, Theorem 3 at $z = 3$ gives $\lceil 21/2 \rceil = 11$ and parity lifts it to 12. The circulant of Section 2 has order 12. Hence $n[3, 1; 4] = 12$. $\square$

At $z = 2$, Theorem 3 returns $\lceil 16/2 \rceil = 8$, the published value $n[2, 1; 4] = 8$ of [1].

Remark (the simplicity hypothesis is load-bearing). Without it the value drops to at most 6. On $\mathbb{Z}_6$ take the arc $i \rightarrow i + 1$ with multiplicity three and the edges $\{i, i + 3\}$: every vertex is the tail of three arcs, the head of three arcs and incident with one edge; parallel arcs point the same way, so they create no cycle of length 2; $3 \notin \{1\}$, so no arc is parallel to an edge; $1 + 1 + 1 = 3 \neq 0 \pmod{6}$, so there is no directed triangle; $1 + 1 = 2 \neq 3$, so there is no mixed triangle; and $0 \rightarrow 1 \rightarrow 2 \rightarrow 3$ with the edge $\{3, 0\}$ is a cycle of length 4. The multigraph variant is not decided here.

4. PRIOR WORK

Delete the perfect matching from a $[z, 1; 4]$ -mixed graph and there remains a digraph of minimum out-degree z with no digon and no directed triangle; counts of that kind are standard: the introduction of [1] records that Behzad, Chartrand and Wall [2] proved the analogous lower bound $\lceil (5z + 4)/2 \rceil$ on the order of a z -regular digraph of girth 4. We claim no novelty for Lemma 2 or for the count in Theorem 3.

We do not claim that the circulant of Section 2 is the unique simple $[3, 1; 4]$ -mixed cage, and we say nothing about $g \neq 4$ or $r \neq 1$.

5. VERIFICATION

The mathematics needs no machine: the order-12 witness is settled by the difference computations of Section 2, and Lemma 2 together with the count in the proof of Theorem 1 excludes every smaller order in two lines. The accompanying program `verify.py` re-derives that in exact integer arithmetic from the link list printed in Section 2, confirms Lemma 2 by exhaustive enumeration of all loopless, digon-free, directed-triangle-free digraphs on at most five vertices, and rechecks the objects exhibited in Sections 2 and 3. Its transcript `verify.output.txt` records 66 passing checks in nine parts; parts 1–5 and 7 are the ones bearing on the statements above, and parts 6, 8 and 9 are arithmetic on constants quoted from the literature and comparisons no longer discussed here. The transcript’s scope note names the quoted constants, and records that nothing there decides the multigraph variant beyond exhibiting the order-6 object, and that nothing there claims the order-12 witness is unique.

REFERENCES

- [1] G. Araujo-Pardo, C. Hernández-Cruz, and J. J. Montellano-Ballesteros, *Mixed cages*, Graphs Combin. **35** (2019), no. 4, 989–999. doi:10.1007/s00373-019-02050-1; arXiv:1702.07255. Quotations above are from the preprint.
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